

Prompt	Continuation (== 6B)	2.7B	1.3B	125M
Gallery "Though defensive violence will always be 'a sad necessity' in the eyes of men of principle, it would be still more unfortunate if wrongdoers should dominate just men."- St. Augustine "A new idea is first	condemned as ridiculous, and then dismissed as trivial, until finally it becomes what everybody knows."- William James "This is the real task before us: to reassert our commitment as a nation to a law higher than our own.	condemned as ridiculous, and then dismissed as trivial, before being accepted as the most important thing in the world."- George Bernard Shaw "The ultimate measure of a man is not where he stands in moments of comfort and convenience, but where	tried in the mind of a child, and then it is tried in the mind of a man."- Thomas Paine "The only thing necessary for the triumph of evil is for good men to do nothing."- Edmund Burke "	and foremost a man's duty to his fellow man. It is a duty to be a man of principle, and to be a man of principle, and to be a man of principle, and to be a man of principle, and to be a
_GPL(crypto_unregister_alg); int crypto_register_template(struct crypto_template *tmpl) { struct crypto_template *q; int err = -EEXIST;	down_write(&crypto_alg_sem); list_for_each_entry(q, &crypto_template_list, list) { if (q == tmpl)	list_for_each_entry(q, &crypto_alg_list, list) { if (tmpl->name && tmpl->name != q->alg.cra_name)	q = kzalloc(sizeof(*q), GFP_KERNEL); if (!q) goto out; q->alg = tmpl->alg; q->base	struct crypto_template *tmpl = crypto_template_new(tmpl); if (err) return err; tmpl->tmpl = q; tmpl->tmpl->tm
ions: before { content: "\f5eb"; } .fa-discord: before { content: "\f392"; } .fa-discourse: before { content: "\f393	,"; } .fa-divide: before { content: "\f529"; } .fa-dizzy: before { content: "\f567"; } .fa-dna: before	,"; } .fa-digg: before { content: "\f391"; } .fa-dochub: before { content: "\f394"; } .fa-docker: before {	,"; } .fa-digg: before { content: "\f96c"; } .fa-dollar-sign: before { content: "\f155"; } .fa-digiter	,"; } .fa-discus: before { content: "\f394"; } .fa-drapal: before { content: "\f395"; } .fa-drapal-discord
new users as an exploration tour and getting started guide, with exercises at the end of each chapter. For more advanced trainees it can be a desktop reference, and a collection of the base knowledge needed to proceed with system and network administration. This	book contains many real life examples derived from the author's experience as a Linux system and network administrator, trainer and consultant. They hope these examples will help you to get a better understanding of the Linux system and that you feel encouraged to try out things on	book is designed to give the reader a firm understanding of the technologies needed to install and manage Linux systems, using the various available tools and techniques for the task. The book begins with a rapid-fire introduction to the basic principles of the Linux operating	is a good place to get started for a new user. A: I would recommend the book "Linux Networking" by David S. It is a very good book for beginners. A: I would recommend	is a great way to get started with a new project. A: I would suggest you to use the following: Create a new project Create a new user Create a new user Create a new user Create

Figure 6: Text examples that are memorized by the 6B model, but not by smaller models. Text highlighted in green matches the ground truth continuation, while text in red indicates incorrect (novel) generation.

C ALTERNATE EXPERIMENTAL SETTINGS

In this section, we study other strategies that we could have used to quantify memorization.

Random dataset sampling. In Section 4.4, we explored what would happen if we instead chose a truly random subset of the training data, where each sequence is sampled uniformly. Specifically, we randomly sample 100,000 sequences from The Pile dataset of length 100, 200, 500, and 1,000; prompt the model with the first $N - 50$ tokens; and then test for memorization by verifying if the model can emit the remaining 50 tokens perfectly. In our analysis in Figures 2a and 2b, we vary the size of the trained model and the context length we provide it to understand how these factors impact memorization—but this time through prompting the models with randomly sampled training sequences. As expected, the absolute probability of memorization is much lower than in Figure 1 where we prompted models with training data from the sampled duplication-normalized subset.

We observe similar trends with model scale and context length as in our other results. Larger models memorize more training examples than smaller models—and much more than the baseline GPT-2 model that was not trained on The Pile. Similarly, providing more context to a model increases the likelihood we can discover memorization. In Figure 2b, we prompt models with: prompt length = sequence length - 50. We see that the longer prompts are easier to predict correctly than shorter prompts. The baseline GPT-2 model is nearly twice as accurate on sequences of length 1,000 (prompt length = 950) compared to sequences of length 100 (prompt length = 50).

Alternate definition of extractability. Our main experiments report a sequence as “extractable” if the model’s generated continuation is identical to the true suffix within that training example. This method is a loose lower bound on memorization. Consider two sequences x_1, x_2 both contained in the training dataset. Suppose these two sequences share the same prefix, and differ only in the final suffix; that is, $x_1 = [p|s_1]$ and $x_2 = [p|s_2]$. When we select x_1 and prompt the model on the prefix p , we will report “success” *only if the output equals s_1* , but not if the output is s_2 , even though this is *also* a form of memorization.

We now consider how our results would change if we instead checked that the generation $[p|f(p)]$ from a prompt p was contained *anywhere* in the training dataset. This gives a strictly larger measurement of memorization. By comparing these two methods (checking for memorization within the ground truth continuation, and within the entire dataset), we can understand how the choice of measurement affects the results in our experiments.

Searching within the entire dataset finds more memorized content than comparing with the ground truth (Figure 2c). For examples at 100 repetitions 32.6% of outputs are contained somewhere in the dataset but just 15.8% match the ground truth continuation. This difference becomes more pronounced as the number of repetitions increases. The maximum difference between these approaches is 28.4%, at 2,200 repetitions.

We refrain from using this approach for our main experiments, because this definition requires vastly larger computation resources; it requires querying whether hundreds of thousands of sequences are contained in an 800GB training dataset. Therefore, to promote reproducibility, the remainder of this paper continues with testing the generated suffix against the single expected training suffix.

D TEXT MEMORIZED BY ONLY SOME MODELS

Table 1: The number of sequences memorized by one model, and not memorized by another.

Model	Memorized	Not Memorized By			
		125M	1.3B	2.7B	6B
125M	4,812	-	328	295	293
1.3B	10,391	5,907	-	1,205	1,001
2.7B	12,148	7,631	2,962	-	1,426
6B	14,792	10,273	5,402	4,070	-

Table 1 shows the total number of sequences that are memorized by one model but not another. Larger models have more uniquely memorized sequences, although every model has some memorization not shared by any other model. (Even the 125M model memorizes a few sequences that the 6B model does not.)

E MEMORIZATION IN OPT MODELS

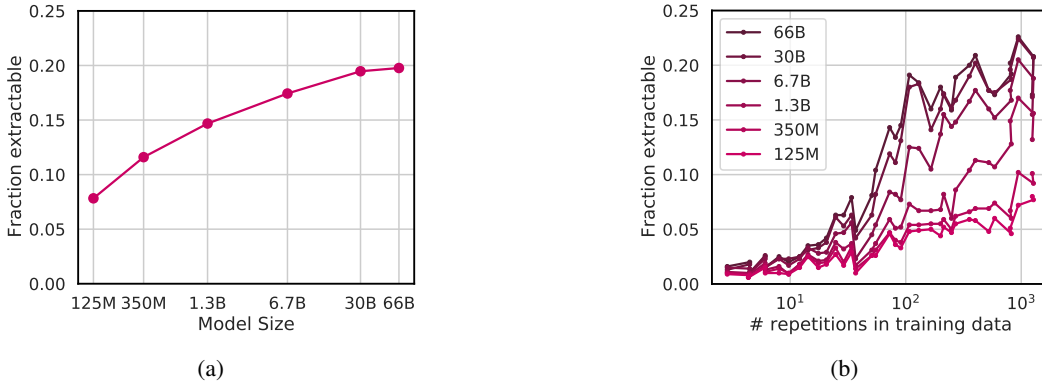


Figure 7: We prompt OPT models with data sampled from their training set. We use a prompt length of 100 here. **(a)** Fraction of sequences extracted as a function of model scale. **(b)** Fraction of sequences extracted as the number of repetitions of that sequence in the training set increases.

F EXAMPLES OF MEMORIZED TEXTS

We show examples of texts that are memorized by different models. We consider the case of 50-token prompts and 50-token generation. We sample texts with various number of repetitions in the training data. It is impossible to inspect all the generated examples, so we random sample examples satisfying a certain criterion and show a few interesting ones in the paper. Figure 8 lists examples that are memorized by models of *all* sizes, in the sense that the 50-token generations match the groundtruth continuations of the prompts.

Prompt	Continuation (== 6B == 2.7B == 1.3B == 125M)
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** This program is free software; you can redistribute it and/or modify * it under the terms of the GNU General Public License version 2 and * only version 2 as published by the Free Software Foundation. **	This program is distributed in the hope that it will be useful, * but WITHOUT ANY WARRANTY; without even the implied warranty of * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the *
Privacy & Cookies Policy Privacy Overview This website uses cookies to improve your experience while you navigate through the website. Out of these cookies, the cookies that are categorized as necessary are stored on your browser as they are essential for the working	of basic functionalities of the website. We also use third-party cookies that help us analyze and understand how you use this website. These cookies will be stored in your browser only with your consent. You also have the option to opt-out of
\end{document}\$ in front of \$\documentclass[12pt]{minimal} \usepackage{amsmath}	\usepackage{wasysym} \usepackage{amsfonts}
Len int for shift := uint(0); ; shift += 7 { if shift >= 64 { return ErrIntOverflowRaft	} if iNdEx >= i { return io.ErrUnexpectedEOF } b := dAtA[
</object> <nil key="sourceID"/> <int key="maxID">18</int> </object> <object class="IBClassDescriber" key="	IBDocument.Classes"> <object class="NSMutableArray" key="referencedPartialClassDescriptions"> <bool key="EncodedWithXMLCoder">YES</bool

Figure 8: Text examples that are memorized by all the models: given 50-token prompts on the left, the next 50 tokens generated by all the models match the groundtruth continuation.

Figure 9 lists examples that are memorized by the 6B model but not by smaller ones. Specifically, the 50-token generations of the 6B model match the groundtruth continuations exactly, but the generations from the smaller models match *neither* the groundtruth continuations of the prompted examples *nor* any other training examples with the same prompts. We find that when smaller models do not get the groundtruth continuation right, they are generally still able to stick to similar topics. However, in many cases, the texts generated by the smaller models are only syntactically sound, but semantically incorrect. Figure 10 and Figure 11 show more examples.

In Figure 12 we show examples that are only memorized by the smallest model, using similar criterion as when we filter examples that are only memorized by the largest model. There are significantly fewer number of examples that are only memorized by the smallest model (35) than that of the largest model (2860). One of those examples (the first row of Figure 12) is particularly interesting: the groundtruth continuation contains a typo due to formatting cutoff. While the smallest model memorized the typo, larger models try to fix the typo.

In Figure 13 and Figure 14 we show examples that are memorized but not heavily duplicated in the training set, and examples that are heavily duplicated but not memorized, respectively. Finally, we show examples that are memorized by GPT2-XL in Figure 15.